

Seminar On Algebraic Geometry
Lecture Notes 1 (Incomplete version)

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1 Sheaves

Grothendieck's creature in algebraic geometry arises mcuh from differential geometry. Roughly speaking, Grothendieck generalized both differential geometry and algebraic geometry, in order to provide a new geomtric view with algebraic objects.

Definition 1.1. Let $(X, \mathcal{T})$ be a topology space. A **presheaf** $\mathscr{F}$ of category $\mathcal{C}$ is a contravarinat functor

$$\mathscr{F} : \mathsf{Top}(X) \to \mathscr{C},$$

where $\mathrm{Top}(X) = (\mathcal{T}, \subseteq)$ is a preorder category.

In other words, if $\mathscr{F}$ is a presheaf of Ab, then for every open subset U of X, there is an abelian group $\mathscr{F}(U)$ and for every inclusion $U \subseteq V$ of open subsets of X, there is a morphism of abelian groups $\rho_{VU} = \mathrm{res}_{V,U} : \mathscr{F}(V) \to \mathscr{F}(U)$. And it satisfies

(1) $\mathscr{F}(\varnothing) = 0$; (Not necessary)

(2) $\rho_{UU} = \mathrm{id}_U : \mathscr{F}(U) \to \mathscr{F}(U)$;

(3) If $U \subseteq V \subseteq W$ are three open subsets, then $\rho_{WU} = \rho_{WV} \circ \rho_{VU}$.

$$\begin{array}{ccc} U & \longrightarrow & \mathscr{F}(U) \\ \big\downarrow & & \big\uparrow{\scriptstyle \rho_{VU}} \\ V & \longrightarrow & \mathscr{F}(V) \end{array}$$

As a matter of terminology, if : $\mathscr{F}$ is a presheaf on X, we refer to: $\mathscr{F}(U)$ as the **sections** of the presheaf $\mathscr{F}$ over the open set U, and we sometimes use the notation $\Gamma(U, \mathscr{F})$ to denote the group $\mathscr{F}(U)$. We call the maps $\rho_{VU} = \mathrm{res}_{V,U}$ **restriction maps**, and we sometimes write $s \mid_V$ instead of $\rho_{VU}(s)$, if $s \in \mathscr{F}(U)$.

We always assume that $\mathcal{C}$ be Ab, i.e., the category of abelian groups, in order to give the category of presheaves the abelian category constructure.

Simply, a presheaf is just a contravariant functor $\mathscr{F} : \mathcal{D} \to \mathcal{C}$. A sheaf is a presheaf whose sections are, in a technical sense, uniquely determined by their restrictions, via "glueing together".

Definition 1.2. A presheaf $\mathscr{F}$ on a topological space X is a **sheaf** if it satisfies the following supplementary conditions:

(4) **Identity axiom:** If U is an open subset, if $\{V_i\}$ is an open covering of U, and if $s \in \mathscr{F}(U)$ is an element such that $s\mid_{V_i} = 0$ for all i, then $s = 0$. It is equivalanet with that $s\mid_{V_i} = t\mid_{V_i} \implies s = t$.

(5) **Glueing axiom:** If U is an open subset, if $\{V_i\}$ is an open covering of U, and if we have elements $s_i \in \mathscr{F}(V_i)$ for each i, with the property that for each i, j, $s_i\mid_{V_i \cap V_j} = s_j\mid_{V_i \cap V_j}$ then there is an (by (4), unique) element $s \in \mathscr{F}(U)$ such that $s\mid_{V_i} = s_i$ for each i.

Example 1.3. [Ha, Ex. II 1.1] Let A be an abelian group, and define the **constant presheaf** $\underline{A}_{\text{pre}}$ associated to A on the topological space X to be the presheaf $U \mapsto A$ for all $U \neq \varnothing$, with restriction maps the identity. In particular, the constant presheaf $\underline{\mathbb{Z}}_{\text{pre}}$ associated to $\mathbb{Z}$ is given by $U \mapsto \mathbb{Z}$, and any section $s \in \underline{\mathbb{Z}}_{\text{pre}}(U)$ is an integer $s \in \mathbb{Z}$.

[Ha, p. II 1.0.3] Let X be a topological space, and A an abelian group. We define the **constant sheaf** $\underline{A}$ on X determined by A as follows.

Give A the discrete topology, and for any open set $U \in X$, let $\underline{A}(U)$ be the group of all continuous maps of U into A:

$$\underline{A}(U) := C(U, A).$$

The restriction maps are the usual restrictions of maps. Since A is a discrete topological ring, each $\underline{A}(U)$ is endowed with ring structure. Immediately, $\underline{A}$ is a ring sheaf on X.

Note that for every connected open set U, $\underline{A}(U) \cong A$, whence the name "constant sheaf". To see this, let $s \in \mathscr{A}(U)$ be a section, then $U = \bigcup_{a \in A} s^{-1}(a)$. Since U is connected, there is only one $s^{-1}(a) = U$ and the others are empty. Thus, we obtain a surjective homomorphism below

$$\mathscr{A}(U) \to A \quad \text{with} \quad s \mapsto s(x),\ \forall x \in U.$$

It is also injective, since $s(U) = \{0\}$ if and only if $s = 0$.

Therefore, if U is an open set whose connected componentes are open, then $\underline{A}(U) = \prod A$, one for each connected components of U.

Here we define sheaves of modules in order to take a more geometric viewpoint on sheaves. They are studied in [Ha, p. II.5].

Definition 1.4. A **ringed space** is a pair $(X, \mathscr{O}_X)$ consisting of a topological space X and a sheaf of rings $\mathscr{O}_X$ on X.

Definition 1.5. Let $(X, \mathscr{O}_X)$ be a ringed space. A **sheaf of $\mathscr{O}_X$-modules** is a sheaf $\mathscr{F}$ on X, such that for each open set $U \subseteq X$, the abelian group $\mathscr{F}(U)$ is an $\mathscr{O}_X$-module, and for for each inclusion of open sets $V \subseteq U$, the restriction homomorphism $\mathscr{F}(U) \to \mathscr{F}(V)$ is compatible with the module structure via the ring homomorphism $\mathscr{O}_X(U) \to \mathscr{O}_X(V)$.

Far from being abstract nonsense, sheaf theory generalizes and unifies many geometric objects. Here are some examples from geometry and analysis.

Example 1.6. Let X be a topological space. For each open subset $U \subseteq X$, let $C(U)$ denote the $\mathbb{R}$-algebra of continuous functions $U \to \mathbb{R}$, with restriction maps given by restriction of functions. Then $\mathscr{C}$, defined by $\mathscr{C}(U) = C(U)$, is a sheaf of $\mathbb{R}$-algebras on X; the glueing axiom follows from the local nature of continuity. In particular, if M is a smooth manifold, then the structure sheaf $\mathscr{O}_M$ is the sheaf of smooth functions, given by

$$\mathscr{O}_M(U) = C^\infty(U).$$

Example 1.7. Let X be a complex manifold. For each open subset $U \subseteq X$, let $\mathcal{O}(U)$ denote the $\mathbb{C}$-algebra of holomorphic functions $U \to \mathbb{C}$, with restriction maps given by restriction of functions. Then the structure sheaf $\mathscr{O}_X$, defined by $\mathscr{O}_X(U) = \mathcal{O}(U)$, is a sheaf of $\mathbb{C}$-algebras on X: if $\{U_i\}$ is an open cover of U and $f_i \in \mathcal{O}(U_i)$ agree on all overlaps $U_i \cap U_j$, then there is a unique $f \in \mathcal{O}(U)$ restricting to each f_i, since holomorphicity is a local property. This is the holomorphic analogue of the sheaf of smooth functions on a smooth manifold.

Already here one sees why one works with sheaves rather than only global sections: on a compact connected complex manifold X, every global holomorphic function is constant, so $\mathscr{O}_X(X) \cong \mathbb{C}$, whereas the sheaf $\mathscr{O}_X$ still encodes rich local holomorphic data.

Let M be a smooth manifold. For each open subset $U \subseteq M$, let $\Omega^k(U)$ denote the $C^\infty(U)$-module of smooth k-forms on U, with restriction maps given by restriction of forms. Then Ω^k is a sheaf of C^∞_M-modules on M.

Crucially, Ω^k is not merely a sheaf of abelian groups: for $f \in C^\infty_M(U)$ and $\omega \in \Omega^k(U)$, the product $f\omega$ is defined pointwise by $(f\omega)_p = f(p)\omega_p$, and

restriction is compatible with this structure, making Ω^k a sheaf of modules over the structure sheaf $\mathscr{O}_M$. In particular, $\Omega^0 = \mathscr{O}_M$, and Ω^1 is the sheaf of sections of the cotangent bundle T^*M.

Indeed, sheaves of modules provide an analogue of fibre bundles in algebraic geometry. For example, Ω^k is a $\mathscr{O}_M$-module, and is locally free of rank $\binom{n}{k}$ over $\mathscr{O}_M$, hence corresponds to the vector bundle $\bigwedge^k T^*M$, i.e., $\Omega^k = (-, \bigwedge^k T^*M)$.

There is a deep relation between sheaves of modules with bundles. For example, in differential geometry, the Swan-Serre theorem states that: for X a smooth manifold with $\mathbb{R}$-algebra of smooth functions $C^\infty(X)$, there is an equivalence of categories between that of finite rank smooth vector bundles over X and finitely generated projective modules over $C^\infty(X)$. See [Ne, Chapter 12] for more details.

Example 1.8. Let $p : Y \to X$ be a continuous map. For each open subset $U \subseteq X$, define

$$\Gamma(Y/X)(U) = \{s : U \to Y \mid p \circ s = \mathrm{id}_U,\ s \text{ continuous}\},$$

with restriction maps given by $s \mapsto s|_V$. Then $\Gamma(Y/X)$ is a sheaf on X, called the *sheaf of sections* of p.

If $p : Y \to X$ is a vector bundle, then $\Gamma(Y/X)(U)$ is a $C^\infty(U)$-module, and $\Gamma(Y/X)$ is a sheaf of C^∞_X-modules, locally free of rank equal to the rank of the bundle. Indeed, the name "section" comes from such sections in differential geometry.

Example 1.9. Let M be a smooth manifold, for any open subset $U \subseteq X$, define $C^\infty_c(U)$ be the $\mathbb{R}$-vector space of smooth functions from U to $\mathbb{R}$ with compact support. Then C_c forms a presheaf, which is not a sheaf in general.

For example, let $M = \mathbb{R}$, φ be a bump function such that $\mathrm{supp}(\varphi) = [-1/3, 1/3]$, and $\varphi(0) = 1$. Let $f(x) = \sum_{n\in\mathbb{Z}} \varphi(x-n)$. Then $\mathrm{f}(\varphi) = \bigcup_{n\in\mathbb{Z}}[n - 1/3, n + 1/3]$, hence $f \notin C_c(\mathbb{R})$. Now let $U_n := (n - 1/2, n + 2/3)$ be some open subsets covering $\mathbb{R}$. In each U_n, $\mathrm{supp}(f\ |_{U_n}) = [n-1/3, n+1/3]$ which is compact. However, the global section glueing by $f\ |_{U_n}$ is identically f, which is not in $C^\infty_c(\mathbb{R})$.

An interesting result is, as the dual topological space of $C^\infty_c(U)$, the space of distributions $\mathcal{D}'(U)$ forms a sheaf. For a brief introduction to this, see [WkDi, p. 4.5].

Now we come back to classical algebraic geometry. In [Ha, p. I.3], we define the ring of regular functions on an algebraic variety V, denoted by $\mathscr{O}(V)$.

Example 1.10. [Ha, p. II.1.0.1] Let V be an algebraic variety over the field k. For each open set $U \subseteq V$, let $\mathscr{O}_V(U) := \mathscr{O}(U)$ be the ring of regular functions from U to k. Then $\mathscr{O}_V$ is a presheaf of rings on X.

$\mathscr{O}_V$ is a sheaf, which is called the *sheaf of regular functions* on X. This is because the regularity of a function is defined locally.

Now we define the morphisms between presheaves on the same space.

Definition 1.11. For a presheaf $\mathscr{F}$ is a contravariant functor $\mathscr{F} : \mathsf{Top}(X) \to \mathsf{Ab}$, it forms a functor category $[\mathsf{Top}(X)^{\mathsf{op}}, \mathsf{Ab}]$. We define the category of presheaves as $\mathsf{PSh}(X) = [\mathsf{Top}(X)^{\mathrm{op}}, \mathsf{Ab}]$.

Thus, the morphism $\varphi : \mathscr{F} \to \mathscr{G}$ is a natural transformation, which consists of morphisms of abelian groups $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$ for each open set U, such that whenever $V \subseteq U$ is an inclusion, the diagram

$$\begin{array}{ccc} \mathscr{F}(U) & \xrightarrow{\varphi(U)} & \mathscr{G}(U) \\ \downarrow{\scriptstyle \rho_{UV}} & & \downarrow{\scriptstyle \rho_{UV}} \\ \mathscr{F}(V) & \xrightarrow{\varphi(V)} & \mathscr{G}(V) \end{array}$$

commutes. Or we can simply say that $\varphi(s\mid_V) = \varphi(s)\mid_V$, for any section $s \in \mathscr{F}(U)$.

We can simply define the category of sheaves on X to be the full subcategory $\mathsf{Sh}(X)$ of $\mathsf{PSh}(X)$.

Definition 1.12. Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of presheaves. We define the presheaf kernel of φ, presheaf cokernel of φ, and presheaf image of φ to be the presheaves given by $U \mapsto \ker \varphi(U)$, $U \mapsto \operatorname{coker} \varphi(U)$, and $U \mapsto \operatorname{im} \varphi(U)$ respectively.

Indeed, PSh is an abelian category.

Lemma 1.13. If $\varphi : \mathscr{F} \to \mathscr{G}$ is a morphism of sheaves, then the presheaf kernel of φ, $\ker \varphi$ is also a sheaf.

Proof. Notice that $(\ker \varphi)(U) \subseteq \mathscr{F}(U)$ hence we can use the sheaf axiom on $\mathscr{F}$. $\square$

2 Stalks & Germs

Before we investigate the category of sheaves and operations on sheaves, we first introduct the stalks of sheaves.

Definition 2.1 (Stalks)**.** If $\mathscr{F}$ is a presheaf on X, and if P is a point of X, we define the *stalk* $\mathscr{F}_P$ of $\mathscr{F}$ at P to be the direct limit (colimit) of the groups $\mathscr{F}(U)$ for all open sets U containing P, via the restriction maps ρ, as

$$\mathscr{F}_P := \varinjlim \mathscr{F}(U).$$

The direct limit means that, for any open neighborhood U of P, there exists a homomorphism $\rho_U : \mathscr{F}(U) \to \mathscr{F}_P$, such that, for compatible homomorphisms $\lambda_U : \mathscr{F}(U) \to A$, there exists a unique homomorphism $\mathscr{F}_P \to A$.

Elements in $\mathscr{F}_P$ are called *germs* of sections of $\mathscr{F}$ at the point P.

In fact, the index category $\mathsf{Top}_P(X)$ with all open neighborhood U of P is a *filtered category*, since for any open neighborhood U and V of P, $U \cap V$ is also an open neighborhood of P. By [Va, p. 1.4 D], the germ in $\mathscr{F}_P$ has the form of $\langle U, s\rangle$ while U is the open neighborhood of P and $s \in \mathscr{F}(U)$ under the equivalence relation that

$$\langle U, s\rangle = \langle V, t\rangle \iff \exists \text{ an open neighborhood } W \subseteq U \cap V \text{ of } P \text{ s.t. } s\,|_W = t\,|_W .$$

In fact, taking stalks gives a functor $\mathsf{Sh}(X) \to \mathsf{Ab}$, which is exact and left-adjoint.

Example 2.2 (Analytic germs)**.** The name of germs came from analytic continuation. See [Ah, Chapter 8].

Definition 2.3 (Etale space)**.**

3 Local Nature of Sheaves

Lemma 3.1. For every $P \in U$, there is a map $\mathscr{F}(U) \to \mathscr{F}_P$, sending a section $s \in \mathscr{F}(U)$ to the equivalence class of (U, s), denoted s_P.

If $\mathscr{F}$ is a sheaf, a section $s \in \mathscr{F}(U)$ of $\mathscr{F}$ over U is determined by its images in the stalks $\mathscr{F}_x$ for all $x \in U$, i.e. $s = 0$ if and only if $s_P = 0$, for all $P \in U$. Equivalently, $s = t \in \mathscr{F}(U)$ if and only if $s_P = t_P \in \mathscr{F}_P$, for all $P \in U$.

Proof. For any germ (s_P, U_P), such that $S_P = 0$, we can obtain the zero section by glueing s_P, since U_P forms an open cover of X. By the identity axiom, any section glueing by s_P must be zero. □

Proposition 3.2. [Va, 2.4 N] Consider the morphism $\varphi : \mathscr{F} \to \mathscr{G}$ of sheaves on the topology space X. Then, the conditions below are equivalent:

(1) φ is a monomorphism.

(2) φ is injective on the level of stalks: $\varphi_P : \mathscr{F}_P \to \mathscr{G}_P$ is injective for all $P \in X$.

(3) $\ker \varphi = 0$.

(4) φ is injective on the level of open sets : $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$ is a monomorphism of Ab, thus injective and $\ker \varphi(U) = (0)$, for all open set $U \subseteq X$.

Proof. We prove this by (2) $\Longrightarrow$ (1) $\Longrightarrow$ (3) $\Longrightarrow$ (4) $\Longrightarrow$ (2).

(2) $\Longrightarrow$ (1) : By 3.1.

(1) $\Longrightarrow$ (3) : By the definition of $\ker \varphi$ we could obtain an inclusion morphism $i : \ker \varphi \to \mathscr{F}$ and a zero morphism $0 : \ker \varphi \to \mathscr{F}$ sending $(\ker \varphi)(U) \mapsto (0)$ inducing the commutative diagram below

$$\ker \varphi \underset{0}{\overset{i}{\rightrightarrows}} \mathscr{F} \xrightarrow{\varphi} \mathscr{G}.$$

Since φ is a monomorphism, i must be 0, which means that $\ker \varphi = 0$.

(3) $\Longrightarrow$ (4) : Notice that $\ker(\varphi(U)) = (\ker \varphi)(U)$.

(4) $\Longrightarrow$ (2) : If $\ker \varphi(U) = (0)$ for all open sets $U \subseteq X$, then for any $P \in X$, and germs $s_P \in \ker \varphi_P$, there exists an open neighborhood U of P, such that $s_P = \langle U, s \rangle$ for some $s \in \ker \varphi(U)$. By the assumption, $s = 0$, hence $s_P = 0$.

□

Proposition 3.3. [Va, 2.4 O] Consider the morphism $\varphi : \mathscr{F} \to \mathscr{G}$ of sheaves on the topology space X. Then, the conditions below are equivalent:

(1) φ is an epimorphism.

(2) φ is surjective on the level of stalks: $\varphi_P : \mathscr{F}_P \to \mathscr{G}_P$ is an epimorphism, thus surjective and $\operatorname{im} \varphi_P = \mathscr{G}_P$, for all $P \in X$.

(3) For every open set $U \subseteq X$, and for every section $s \in \mathscr{G}(U)$, there is a covering $\{U_i\}$ of U, and sections $t_i \in \mathscr{F}(U_i)$ such that $\varphi(t_i) = s \mid_{U_i}$ for all i.

Proof. We prove (1) ⇒ (2) ⇒ (3) ⇒ (1).

(1) ⇒ (2): Assume φ is an epimorphism. Let $\operatorname{Im}\varphi$ be the image sheaf of φ, so that φ factors as

$$\mathscr{F} \xrightarrow{q} \operatorname{Im}\varphi \xrightarrow{i} \mathscr{G},$$

where i is a monomorphism. Since φ is an epimorphism, i is also an epimorphism: indeed, if $\alpha\circ i = \beta\circ i$, then $\alpha\circ\varphi = \beta\circ\varphi$, hence $\alpha = \beta$. Thus i is both a monomorphism and an epimorphism, hence an isomorphism. Taking stalks gives

$$\operatorname{im}\big(\varphi_P : \mathscr{F}_P \to \mathscr{G}_P\big) = (\operatorname{Im}\varphi)_P = \mathscr{G}_P,$$

so φ_P is surjective for every $P \in X$.

(2) ⇒ (3): Let $U \subseteq X$ be open and $s \in \mathscr{G}(U)$. For each $P \in U$, the surjectivity of φ_P gives $t_P \in \mathscr{F}_P$ with $\varphi_P(t_P) = s_P$. Choose a representative $t^P \in \mathscr{F}(V_P)$ of t_P on some open neighbourhood $V_P \subseteq U$ of P. Then $\varphi(t^P)$ and s have the same stalk at P, so there is an open neighbourhood $W_P \subseteq V_P$ of P such that

$$\varphi(t^P|_{W_P}) = s|_{W_P}.$$

The sets $\{W_P\}_{P\in U}$ cover U, and the sections $t^P|_{W_P} \in \mathscr{F}(W_P)$ satisfy the required condition.

(3) ⇒ (1): Let $\alpha, \beta : \mathscr{G} \to \mathscr{H}$ be morphisms of sheaves with $\alpha \circ \varphi = \beta \circ \varphi$. For any open $U \subseteq X$ and any $s \in \mathscr{G}(U)$, condition (3) gives an open cover $\{U_i\}$ of U and sections $t_i \in \mathscr{F}(U_i)$ such that $\varphi(t_i) = s|_{U_i}$. Then

$$\alpha(s)|_{U_i} = \alpha(\varphi(t_i)) = \beta(\varphi(t_i)) = \beta(s)|_{U_i}.$$

Since the U_i cover U, we get $\alpha(s) = \beta(s)$. Hence $\alpha = \beta$, so φ is an epimorphism.

Thus (1) ⇔ (2) ⇔ (3). □

Proposition 3.4 (1.1)**.**

4 Sheafification

5 Historical Remark

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